

1. CORE PRODUCT DEFINITION

Objective:

Deliver a personalized, real-time ranked feed optimizing:

- Engagement (CTR, dwell time, watch time)
- Retention (session + long-term)
- Conversion (orders, clicks, purchases)
- Diversity and fairness
- Latency < 150 ms

Key Constraints:

- Strict latency budgets
- High availability
- Real-time adaptability
- Scalability
- Training-serving consistency

2. HIGH-LEVEL SYSTEM ARCHITECTURE

Client → Edge → API Gateway → Feed Orchestrator → Retrieval → Ranking → Cache → Response

Parallel Data Flow:

User Events → Kafka → Flink → Feature Store → Data Lake → Training → Deployment

3. CLIENT AND EDGE LAYER

Components:

- Next.js / Native apps
- CDN (Cloudflare / Akamai)
- Edge functions

Responsibilities:

- Session initialization
- Request compression
- Geo/device tagging
- Latency reduction

4. API GATEWAY AND ORCHESTRATOR

API Gateway:

- Authentication
- Rate limiting
- Routing

Feed Orchestrator:

- Parallel retrieval calls
- Feature coordination
- Latency enforcement
- Fallback handling

5. REAL-TIME FEATURE SYSTEM

Online Feature Store (Redis):

- User embeddings
- Session data
- Real-time counters

Offline Feature Store (Delta Lake / Iceberg):

- Historical data
- Training datasets

Feature Pipelines:

Streaming (Flink):

- Session features
- Trend detection

Batch (Spark):

- Long-term embeddings
- Aggregations

6. EVENT STREAMING SYSTEM

Kafka:

- Impression, click, dwell, skip, purchase, search, session events

Flink:

- Feature updates
- Trend detection
- Anomaly detection

7. CANDIDATE GENERATION

Sources:

- Embedding retrieval (FAISS / Milvus)
- Graph retrieval (collaborative filtering)
- Content-based retrieval
- Trending pool
- Context-aware retrieval
- Search hybrid (BM25 + vector)

Output: ~2000 candidates

8. PRE-FILTERING

- Availability
- Geo constraints
- Content safety
- User exclusions
- Deduplication

9. MULTI-STAGE RANKING

Stage 1:

- GBDT / small NN
- 2000 → 300

Stage 2:

- Transformer + DLRM

Inputs:

- User embeddings
- Item embeddings
- Session sequences
- Context

Outputs:

- CTR
- Dwell
- Conversion
- Retention

Stage 3:

- Listwise reranking
- Diversity, fairness, exploration

10. POLICY LAYER

- Sponsored blending
- Creator fairness
- Fatigue control
- Regulatory filters
- Abuse prevention

11. SERVING INFRASTRUCTURE

- Triton inference server
- ONNX / TensorRT

Latency:

- Feature: 20 ms
- Retrieval: 30 ms
- Ranking: 50 ms
- Rerank: 20 ms

12. CACHING

Redis:

- Hot feeds
- Repeated requests
- Fallbacks

13. CONTINUOUS LEARNING LOOP

User → Kafka → Flink → Feature Store → Data Lake → Training → Deployment → Serving

14. TRAINING PIPELINE

- Kubeflow / Airflow
- PyTorch + Ray
- Negative sampling
- Debiasing

Metrics:

- AUC
- NDCG
- Slice metrics

15. EXPERIMENTATION

- A/B testing
- Metrics:
- CTR
- Retention
- Revenue
- Session length

16. OBSERVABILITY

- Prometheus
- Grafana
- OpenTelemetry

ML Monitoring:

- Drift detection
- KPI tracking

17. FAILURE HANDLING

- Model failure → cached feed
- Retrieval failure → trending
- Feature failure → static fallback

System must never return empty feed

18. SECURITY

- Authentication
- Privacy controls
- Abuse detection

- Content moderation

19. INFRASTRUCTURE

- Docker
- Kubernetes
- Helm
- ArgoCD
- Terraform

Capabilities:

- Autoscaling
- Rolling updates
- Canary deployments

20. CORE INSIGHT

The system is not just the model.

It is:

- Data flow
- Feature consistency
- Latency control
- Feedback loops
- Reliability engineering
- Product constraints